Name

Class

R. No.

- Mathematics
- +1 CBSE/ IITJEE
- Test 7 (September 04,2010)
- Trigonometry
- Maximum Marks 100
- Time 80 minutes

Instructions

1. **The test contains 10 questions.**

Prove the following identities (Q1-Q9) $9 \times 10\ Marks$

Q1: $\dfrac{(\sin 2\alpha - \sin 6\alpha) + (\cos 2\alpha - \cos 6\alpha)}{\sin 4\alpha - \cos 4\alpha} = 2\sin 2\alpha$

Q2: $\cos^2\alpha - \sin^2 2\alpha = \cos^2\alpha\cos 2\alpha - 2\sin^2\alpha\cos^2\alpha$

Q3: $$\frac{1-2\cos^2\alpha}{2\tan\left(\alpha-\frac{\pi}{4}\right)\sin^2\left(\frac{\pi}{4}+\alpha\right)}=1$$

Q4: $1+2\cos 7a=\frac{\sin 10.5a}{\sin 3.5a}$

Q5: $\dfrac{\sqrt{1+\cos\alpha}+\sqrt{1-\cos\alpha}}{\sqrt{1+\cos\alpha}-\sqrt{1-\cos\alpha}}=\cot\left(\dfrac{\alpha}{2}+\dfrac{\pi}{4}\right)$, $\pi<\alpha<2\pi$

Q6: $2\left(\sin^6\alpha+\cos^6\alpha\right)-3\left(\sin^4\alpha+\cos^4\alpha\right)+1=0$

Q7: $\dfrac{1+\tan\alpha+\tan^2\alpha}{1+\cot\alpha+\cot^2\alpha}=\tan^2\alpha$

Q8: $\dfrac{\sqrt{2}-\sin\alpha-\cos\alpha}{\sin\alpha-\cos\alpha}=\tan\left(\dfrac{\alpha}{2}-\dfrac{\pi}{8}\right)$

Q9: $\frac{2\cos^2 2\alpha+\sqrt{3}\sin 4\alpha-1}{2\sin^2 2\alpha+\sqrt{3}\sin 4\alpha-1}=\frac{\sin\left(4\alpha+30^o\right)}{\sin\left(4\alpha-30^o\right)}$

Q10: Simplify the following expressions $9 \times 10\ Marks$

a) $\dfrac{1+\sin 2x}{(\sin x+\cos x)^2}$

b) $\cos 0+\cos\frac{\pi}{7}+\cos\frac{2\pi}{7}+\cos\frac{3\pi}{7}+\cos\frac{4\pi}{7}+\cos\frac{5\pi}{7}+\cos\frac{6\pi}{7}$